

National Taiwan Ocean University
ICPC Codebook

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1. Misc

1.1. Truly Misc

1.1.1. Macros

```
1 #define _GLIBCXX_DEBUG 1
2 #pragma GCC optimize("O3", "unroll-loops")
3 #pragma GCC optimize("fast-math")
4
```

1.1.2. <random>

```
1 #include <random>
2 #ifdef __unix__
3 std::random_device rd;
4 std::mt19937_64 RNG(rd());
5 #else
6 const auto SEED = chrono::high_resolution_clock::now()
7     .time_since_epoch()
8     .count();
9 std::mt19937_64 RNG(SEED);
10 #endif
11
```

2. Data Structure

2.1. 2DBIT.cpp

```
1 #include <bits/stdc++.h>
2 using ll = long long;
3 #define pii pair<int, int>
4 #define pll pair<ll, ll>
5 #define N 1005
6 #define F first
7 #define S second
8 #define lb(x) (x & -x)
9 using namespace std;
10 int bit[N][N];
11 void upd(int i, int j, int v)
12 {
13     for(; j < N; j += lb(j))
14         for(int k = i; k < N; k += lb(k)) bit[k][j] += v;
15 }
16 int qry2(int i, int j)
17 {
18     int ans = 0;
19     for(; j; j += lb(j))
20         for(int k = i; k < N; k += lb(k)) ans += bit[k][j];
21     return ans;
22 }
23 int qry(int y1, int x1, int y2, int x2)
24 {
25     return qry2(y2, x2) - qry2(y2, x1 - 1) - qry2(y1 - 1,
26 x2) + qry2(y1 - 1, x1 - 1);
27 }
28 int main()
29 {
30     // REMEMBER TO DO MEMSET
31     memset(bit, 0, sizeof(bit));
32     int n, q, i = 1, j, y, x;
33 }
```

3. Demo

3.1. Demo code

```
1 #include <iostream>
2
3 int main()
4 {
5     int n;
6     std::cin >> n;
7     for(int i = 0; i < n; i++)
8     {
9         std::cout << "Hello NTOU!\n";
10     }
11 }
12
```